

# Kuwait University



جامعة الكويت  
KUWAIT UNIVERSITY

## Project 1: Intelligent Web Crawler Using Heuristic Search

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# Heuristic Search

## Greedy Best-First Search

Search engines in general use very efficient search algorithms to perform and deliver search results in the most accurate and efficient way. The Greedy Best-First Search algorithm is one of the popular heuristic algorithms used in search. Figure1 [1] below show the general flow diagram for GBFS.

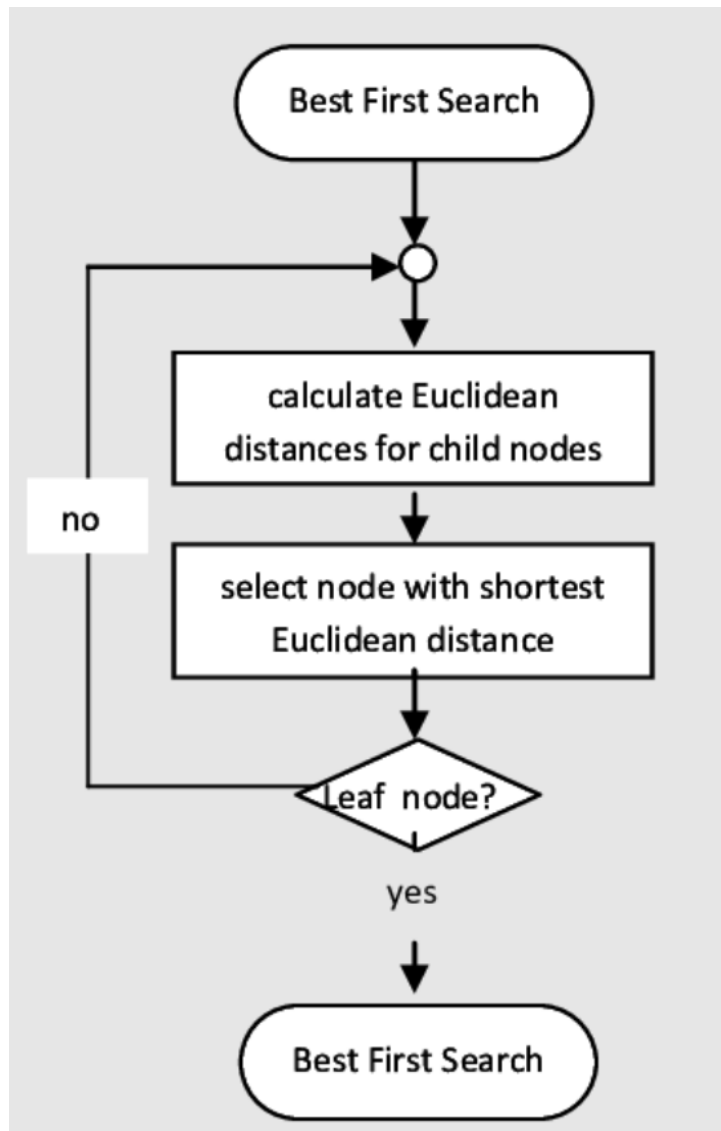


Figure 1: General GBFS Flow

Our approach was to build the graph first; for testing the speed of the algorithm, we have cached the graph by making the nested requests first and cache it by saving it in a 'pickle' file, then build the heuristic function given the keyword. After that it will set the heuristic score based on 4 factors, below the function is the heuristic function that

calculated the score for the GBFS:

$$\text{Score} = 5 \times (\text{occurrences in link text}) \quad (1)$$

$$+ 3 \times (\text{occurrences in surrounding paragraph}) \quad (2)$$

$$+ 2 \times (\text{occurrences in body of HTML}) \quad (3)$$

$$+ 1 \times (\text{links containing similar keywords}). \quad (4)$$

## A\* Search

The heuristic function for our A\* search is as follows:

$$\text{Score} = 4 \times (\text{occurrences in link text}) \quad (5)$$

$$+ 2 \times (\text{occurrences in surrounding paragraph}) \quad (6)$$

$$+ 1 \times (\text{links containing similar keywords}). \quad (7)$$

$$- 75 \times (\text{depth}) \quad (8)$$

Our approach is to treat each page as a node and each link as an edge. The crawler then assigns a value to each link on the page based on its relevance as per the heuristic function. The heuristic function is informed based on both the target phrase and a topic list of words that help it to find the target phrase. We heavily penalized depth to avoid an overtly touristy crawler, since without it the crawler would trend towards depth instead of checking other shallower options.

## Comparison

### GBFS vs. A\*

The A\* algorithm is different from the GBFS in terms of efficiency and performance.

| Search Type | Time                                           | Similar Nodes | Depth |
|-------------|------------------------------------------------|---------------|-------|
| A*          | 8.79 seconds (including requests)              | 1             | 1     |
| GBFS        | 2.2253 seconds (graph is cached - No requests) | 7             | 3     |

Table 1: Comparison Table for link: <https://www.cs.ku.edu.kw> and keyword: alduaij

| Search Type | Time                                                       | Similar Nodes | Depth |
|-------------|------------------------------------------------------------|---------------|-------|
| A*          | 8.79 seconds (including requests)                          | 1             | 1     |
| GBFS        | 76.8825 seconds (graph is not cached - including requests) | 4             | 2     |

Table 2: Comparison Table for link: <https://www.cs.ku.edu.kw> and keyword: alduaij

```

abeer@Mac webCrawling % python3 main.py
~~~~~
To run the greedy best-first search, type '1'
To run A* search, type '2'
To exit, type 'exit'
~~~~~> 1
Enter the start node: https://www.cs.ku.edu.kw
Enter the goal node (can be a description): alduaij

Top similar nodes based on heuristic scores out of 7 similar nodes:
Node: https://nalduaij.com, Score: 54
Node: https://www.cs.ku.edu.kw/research, Score: 49
Node: https://www.cs.ku.edu.kw/faculty, Score: 49
Greedy Best-First Search took 2.2980 seconds.
Path from https://www.cs.ku.edu.kw to alduaij: ['https://www.cs.ku.edu.kw', 'https://www.cs.ku.edu.kw/faculty', 'https://www.cs.ku.edu.kw/naser-
alduaij', 'https://nalduaij.com']
Resulting path as graph edges: [('https://www.cs.ku.edu.kw', 'https://www.cs.ku.edu.kw/faculty'), ('https://www.cs.ku.edu.kw/faculty', 'https://
www.cs.ku.edu.kw/naser-alduaij'), ('https://www.cs.ku.edu.kw/naser-alduaij', 'https://nalduaij.com')]
~~~~~
To run the greedy best-first search, type '1'
To run A* search, type '2'
To exit, type 'exit'
~~~~~> exit

```

Figure 2: Sample Output for GBFS

## Example and Visualization

### Sample run for GBFS

Figure2 is the sample output for the Greedy Best-First Search. The program asks for the type of search weather GBFS or A\* and asks for the start node, which is the source link and the keyword, and then runs the chosen search algorithm.

### GBFS Visualization

In the graph visualization, we can see that there are 2 nodes between the start node and the end node.

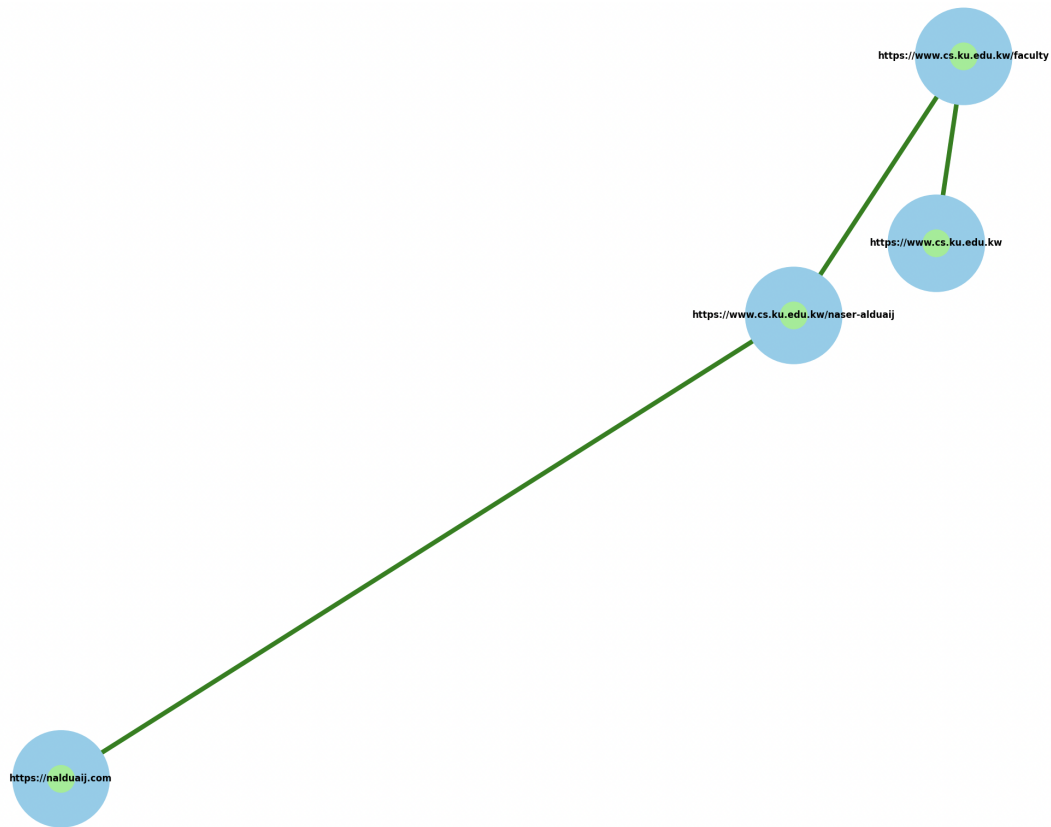


Figure 3: Graph Representation in Visualization

## Sample run for A\*

```

PS C:\Users\foith\Documents\Uni\Advanced AI\PI\webCrawling> python main.py
=====
To run the greedy best-first search, type '1'
To run A* search, type '2'
To exit, type 'exit'
=====
> 2
=====
You have chosen A* search.
Enter the start node: https://www.cs.ku.edu.kw/
Enter the goal node (can be a description): Mohammed A. Almula
=== A* Web Crawler ===
Start web address : https://www.cs.ku.edu.kw/
Target phrase     : 'Mohammed A. Almula'
Topic keywords    : ['computer', 'science', 'kuwait', 'university', 'faculty', 'research', 'capstone', 'exam', 'time', 'people', 'alumni', 'مبحث']

[A*] Exploring page 1: https://www.cs.ku.edu.kw/
[A*] Exploring page 2: https://www.cs.ku.edu.kw/undergraduate/enrollment-data/
[A*] Exploring page 3: https://www.cs.ku.edu.kw/graduate-graduate-program/
[A*] Exploring page 4: https://www.cs.ku.edu.kw/people/teaching-assistants/
[A*] Exploring page 5: https://www.cs.ku.edu.kw/undergraduate/courses-ar/
[A*] Exploring page 6: https://www.cs.ku.edu.kw/graduate/graduate-courses/
[A*] Exploring page 7: https://www.cs.ku.edu.kw/undergraduate/undergraduate-calendar/
[A*] Exploring page 8: https://www.cs.ku.edu.kw/undergraduate/accreditation/
[A*] Exploring page 9: https://www.cs.ku.edu.kw/graduate/
[A*] Exploring page 10: https://www.cs.ku.edu.kw/graduate-schedules/
[A*] Exploring page 11: https://www.cs.ku.edu.kw/computer-labs-booking-procedure/
[A*] Exploring page 12: https://www.cs.ku.edu.kw/research/

Target phrase FOUND at: https://www.cs.ku.edu.kw/research/
Stop reason           : goal_found
Pages visited         : 12
Time taken            : 7.12 seconds

A* path from seed to target:
https://www.cs.ku.edu.kw/
https://www.cs.ku.edu.kw/research/
['https://www.cs.ku.edu.kw/', 'https://www.cs.ku.edu.kw/research/']
=====
To run the greedy best-first search, type '1'
To run A* search, type '2'
To exit, type 'exit'
=====
> exit
=====

```

Figure 4: Sample Output for A\*

## Discussion

From table 1 and table 2, we can see that caching is very important in search algorithms. In table 1 the GBFS has used the cached graph of requested links and represent it as a graph, and once the user give it the keyword the heuristic function will calculate the weights and then the GBFS will give the search result in better time than the A\*. However, in table 2, the performance of the GBFS has declined because the graph was not cached. In addition, the representations of the data structure where the links are used are not the same.

GBFS has a bonus, despite the long time it takes (in case not cached) it has more depth and more similar nodes.

## References

- [1] Faied, Mariam and Mostafa, Ahmed and Girard, Anouck. (2010). Vehicle Routing Problem Instances: Application to Multi-UAV Mission Planning. AIAA Guidance, Navigation, and Control Conference. 10.2514/6.2010-8435.
- [2] GeeksforGeeks, “*Greedy Best-First Search in AI*,” Available at: <https://www.geeksforgeeks.org/artificial-intelligence/greedy-best-first-search-in-ai/>, Accessed on November 4, 2025.
- [3] A. Nigam, “Web crawling algorithms,” *International Journal of Computer Science and Artificial Intelligence*, vol. 4, no. 3, pp. 63–67, Sep. 2014, doi:10.5963/ijcsai0403001.